

MINI ASSESSMENT

Multi-Step Prompt Workflow Using Prompt Chaining

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DEPT: B.TECH CSE

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Topic: Resume Analysis and Job-Match Evaluation

Objective

To design a multi-step prompt workflow using an AI interface without writing code. The workflow breaks a complex resume-analysis task into three smaller steps: **Resume** → **Extract Skills** → **Determine Job Match** → **Generate Final Report**.

Prompt chaining is used so that the output of one step becomes the input for the next step. This makes the overall task easier to organize and produces a consistent final result.

Workflow Design

Step	Task	Prompting Technique
Step 1	Extract Skills and Qualifications	Few-shot prompting
Step 2	Determine Job Match	Step-by-step reasoning
Step 3	Generate Final Result	Structured JSON formatting

Final workflow: RESUME → STEP 1: EXTRACT → EXTRACTED SKILLS → STEP 2: REASON → JOB-MATCH LEVEL → STEP 3: FORMAT → FINAL JSON

Step 1 — Extract Skills and Qualifications

Prompting Technique: Few-shot prompting

Prompt

You are a resume information extractor.

Task: Extract the important skills, qualifications, technologies, and experience areas from the resume.

Return the information as a simple bullet list.

Example 1:

Resume: A candidate knows Python, SQL, Git, and has experience developing web applications.

Skills:

Python

SQL

Git

Web development

Example 2:

Resume: A candidate has Java, Spring Boot, MySQL, and knowledge of REST APIs.

Skills:

Java

Spring Boot

MySQL

REST APIs

Now extract the key skills and qualifications from this resume:

{{RESUME}}

Output only the bullet list of extracted skills and qualifications.

Purpose

This step identifies the important information contained in the resume. Examples are provided to guide the AI toward the expected extraction pattern. This is few-shot prompting because multiple examples establish the desired output format.

Step 2 — Determine Job Match

Prompting Technique: Step-by-step reasoning

Prompt

You are a senior recruitment and job-matching analyst.

Task: Determine how well the candidate matches the given job description using the extracted skills from Step 1.

Possible match levels:

low
medium
high

Guidelines:

Low: Few required skills or qualifications are present.

Medium: Some important requirements are present, but several gaps exist.

High: Most important skills and qualifications match the job requirements.

Step 1 extracted skills:

{{STEP_1_OUTPUT}}

Job description:

{{JOB_DESCRIPTION}}

Compare the candidate's skills with the job requirements step by step and determine the match level.

End with:

Match: low/medium/high

Purpose

This step compares the information extracted from the resume with the requirements of the target job. The output from Step 1 is passed into Step 2 instead of using fixed information, demonstrating prompt chaining.

Step 3 — Generate Final JSON

Prompting Technique: Structured JSON formatting

Prompt

You are a recruitment reporting assistant.

Create the final result using ONLY the information provided below.

Step 1 – Extracted skills:

{{STEP_1_OUTPUT}}

Step 2 – Job-match analysis:

{{STEP_2_OUTPUT}}

Return ONLY valid JSON.

The JSON must contain exactly these three keys:

```
{
  "skills": ["skill1", "skill2"],
  "match": "low/medium/high",
  "summary": "one-sentence summary of the candidate's job match"
}
```

Rules:

"skills" must be an array of strings.

"match" must be exactly "low", "medium", or "high".

"summary" must be one concise sentence.

Do not add any other keys.

Do not use Markdown.

Do not add explanations outside the JSON.

Purpose

The final step converts the information from the previous steps into a structured JSON object. Structured output provides a consistent format for reporting and reliable processing.

End-to-End Test

Input Resume

A candidate has completed a B.Tech in Computer Science and Engineering. The candidate has knowledge of Python, Java, SQL, HTML, CSS, Git, and REST APIs. The candidate has completed a web application project using Python and SQL and has basic knowledge of cloud computing.

Job Description

The company is looking for a junior software developer with knowledge of Python, SQL, REST APIs, Git, web development, and basic cloud concepts.

Step 1 Output — Extracted Skills

- B.Tech in Computer Science and Engineering
- Python
- Java
- SQL
- HTML
- CSS
- Git
- REST APIs
- Web development
- Cloud computing

Step 2 Output — Job Match

The candidate has most of the important skills required for the junior software developer role. Python, SQL, REST APIs, Git, web development, and basic cloud knowledge directly match the job requirements. Therefore, the candidate has a high job match.

Match: high

Step 3 Output — Final JSON

```
{  
  "skills": ["Python", "Java", "SQL", "HTML", "CSS", "Git", "REST APIs", "Web development",  
    "Cloud computing"],  
  "match": "high",  
  "summary": "The candidate has most of the technical skills and qualifications required for the  
    junior software developer role."  
}
```

Final Workflow

RESUME

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STEP 1: EXTRACT

Few-shot prompting

Extract skills and qualifications

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Extracted Skills

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STEP 2: REASON

Step-by-step reasoning

Compare skills with job requirements

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Job-Match Level

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STEP 3: FORMAT

Structured JSON

Generate final result

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FINAL JSON

Conclusion

The designed workflow demonstrates prompt chaining by dividing the resume-analysis task into three smaller and more reliable prompts. Step 1 extracts skills and qualifications using few-shot prompting, Step 2 determines the job-match level using step-by-step reasoning, and Step 3 produces a structured JSON output. The output of each step is passed to the next step, creating a complete end-to-end workflow.

This workflow follows the required **Extract** → **Reason** → **Format** design pattern and demonstrates how an AI interface can be used to solve a multi-step task without writing code.

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